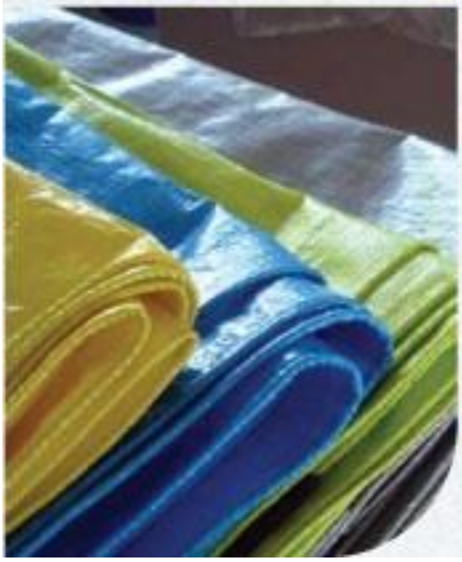
A GUIDE TO PRODUCT QUALITY STANDARDS

PLASTIC PRODUCTS



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ABBREVATIONS

ABCB	Australia Building Codes Board
AICIS	Australian Industrial Chemicals Introduction Scheme
ANSI	American National Standards Institute
APLAC	Asia Pacific Laboratory Accreditation Cooperation
AS	Australian Standards
ASTM	American Society for Testing and Materials
BIS	Bureau of Indian Standards
CE	Conformité Européenne
CoC	Certificate of Conformity
CPCB	Central Pollution Control Board
CPSC	Consumer Products Safety Commission
CPSIA	Consumer Product Safety Improvement Act
CPVC	Chlorinated Polyvinyl Chloride
CPSA	Consumer Product Safety Act
EC	Council Regulation (Regulation of the European Parliament and Council)
ECHA	European Chemical Agency
EN	European Standards
EU	European Union
FCM	Food Contact Materials
FDA	Food and Drug Administration
FD & C	Federal Food, Drug and Cosmetic Act
FHSA	Federal Hazardous Substances Act
FSSAI	Food Safety and Standards Authority of India
GCC	Gulf Co-operation Council
GPSD	General Product Safety Directive
GSO	GCC Standardization Organization
IS	Indian Standard
ISI	Indian Standards Institution
ILAC	International Laboratory Accreditation Co-operation
ISO	International Organization for Standardization
MLA	Multilateral Recognition Arrangement

MRA	Mutual Recognition Arrangement
MSME	Micro, Small & Medium Enterprises
NABCB	National Accreditation Board for Certification Bodies
NABL	National Accreditation Board for Testing and Calibration Laboratories
PVC	Polyvinyl Chloride
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
RoHS	Restriction of Hazardous Substances Directive
SASO	Saudi Standards, Metrology and Quality Organization
SPCB	State Pollution Control Board
UPVC	Unplasticized Polyvinyl Chloride
QCI	Quality Council of India
QCO	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2, 40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This compendium brings relevant regulations and standards pertaining to the plastic industry and its various products. In recent times, several districts in Uttar Pradesh are fast becoming a hub for the production of plastic products including disposable cups, plates, cutlery, tarpaulins, pipes, etc.

PLASTIC PACKAGING, PLATES, CUPS, KANPUR DEHAT

1. INTRODUCTION

Plastic products can be found everywhere owing to their advantages including they being light, possess a high degree of stiffness, ease of production, ability to be produced in high quantities quickly, and high quality. Plastic tableware like plates, cups, spoons, etc. are used in our day- to- day life. Various packaging plastic containers for restaurant take away are common nowadays. Plastic goods industry is rapidly growing in Kanpur Dehat which is a major hub for the production of plastic products including the disposables.

A plastic disposable cup or plate (also called disposable product) is a product designed for a single use after which it is recycled or is disposed as solid waste. The term is also sometimes used for products that may last several months (e.g., disposable air filters) to distinguish from similar products that last indefinitely. Disposable plastic tableware includes:

- Disposable Cups
- Plates
- Plastic Cutlery, etc.

These products are prevalent in fast food restaurants, takeaways, but also for airline meals. In private settings, this kind of disposable products have proven very popular with consumers who prefer easy and quick cleanup after parties without the hassle of having to clean the used utensils. As is the case for disposable cups, materials used are usually plastic (including expanded polystyrene foam), or plastic-coated paper. Recycling rates are especially low for these products, especially when soiled with (wet and / or oily) scraps due to diminished recycled quality.

Food packaging comprises disposable products often found in fast food restaurants, takeout restaurants and kiosks, and catering establishments. They are made from food grade plastics similar to Food-serving items for picnics and parties. Typical disposable food service products are foam food containers, plates, and bowls, cups, utensils, doilies and tray papers. These products can be made from a number of materials including plastics. Plastics are used because the material is lightweight and retains the temperature of hot/cold food and beverages. Foamed polystyrene (sometimes referred to as Styrofoam) is in one of the most common types of plastics used for foodservice packaging in the form of foam food container. Non-foamed polystyrene is sometimes also used for utensils or plastic plates. Polyethylene and other plastics are also used.

2. RAW MATERIALS DISPOSABLE/ SINGLE USE PLASTICS

To make disposable/ single use plastic products, the raw materials used are:

- Polystyrene sheets
- Polypropylene sheets

3. MANUFACTURING PROCESS DISPOSABLE/SINGLE USE CUPS, PLATES & PACKAGING

The manufacturing process is as follows:

3.1. Disposable/Single use Plastic Cups, Plates and Packaging

The disposable plastic cups, plates and packaging are manufactured primarily by thermoforming process:

a. Thermoforming process: Thermoforming process is a manufacturing process used for manufacturing plastic products from thin heated plastic sheets and includes following steps:

- **Heating the plastic film:** The plastic film, approximately 2mm thick is heated to make it pliable in a specially designed oven.
- **Stretching over the mould:** The plastic film, is stretched over the moulds and using conveyor belts is transported to the cutting station, where a die cuts the extruded plastic and trims it.
- **Cooling:** The plastic product is cooled and allowed to firm into its shape. The items thus formed are stocked and the punched waster sheet is wound on scrap sheet winder. To get printed designs on the cups, plates etc., the sheets are printed before forming into cup. This process however generates a sizeable amount of wastage.

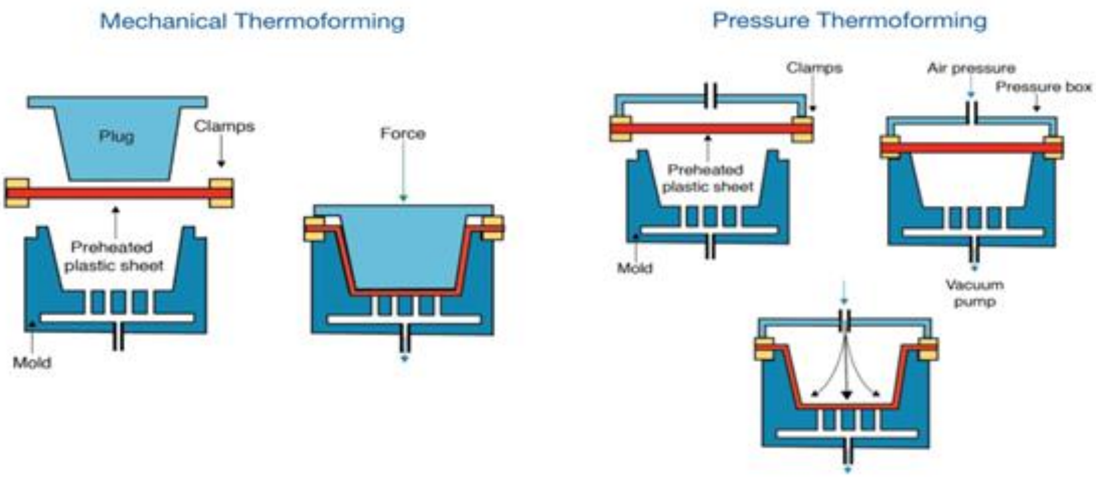

Figure 1: Typical process flow, Thermoforming

3.2. General Quality Checks

Following are a few visual quality parameters/checks:

- Check for cuts or tears in the products formed
- The logo's or designs should be printed properly on the body of the formed product
- Check for any product uniformity issues, including appearance, volume, weight and geometric measurement

This process is viable only however when the products to be formed are thin in their cross section like disposable cups, plates, food packaging used by restaurants etc. This process also generates a large amount of scrap plastic.

4. RAW MATERIALS REUSEABLE PLASTICS

To make reusable plastic plates, cups and other products, the raw materials used are:

- High Density Polyethylene (HDPE)
- Low Density Polyethylene (LDPE)
- Polyethylene Terephthalate (PET)

5. MANUFACTURING PROCESS FOR REUSABLE CUPS, PLATES & PACKAGING

The manufacturing process is as follows:

5.1. Reusable Plastic Cups, Plates and Packaging

The reusable plastic cups, plates and packaging are manufactured primarily by the following process:

a. **Injection Moulding:** Injection moulding is process of manufacturing plastic products at a very rapid rate. This is a cost intensive process and only feasible where large production volumes offset the steep initial costs for manufacturing of the moulds and machinery capex investment. Various steps involved in the injection moulding process are as follows:

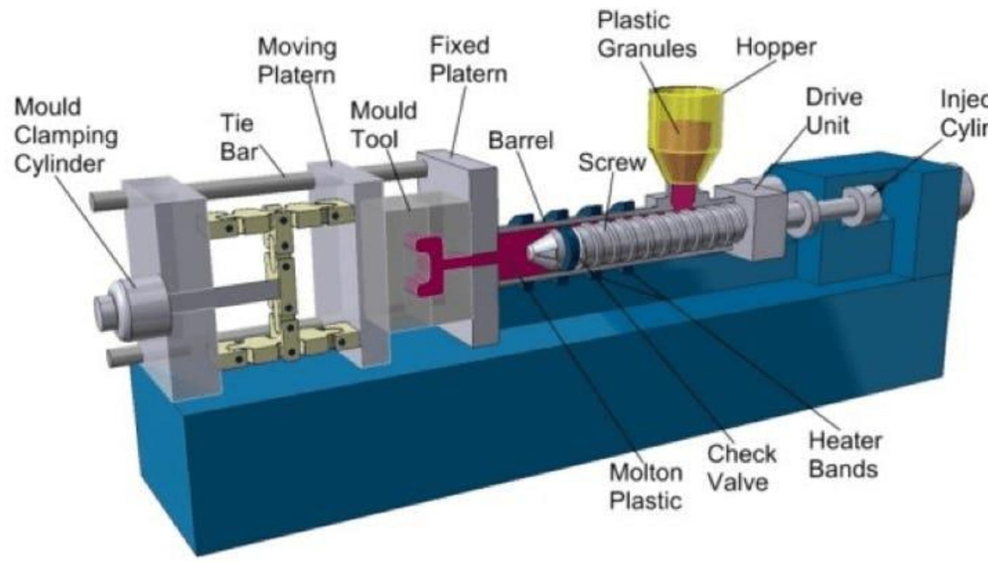

Figure 2: Typical Flow Process for Injection Moulding

- **Mould and mould design**: The choice of materials to manufacture the moulds depends largely on the economics of the operation. Steel moulds are the most expensive to make but their longer life span offsets the high initial cost while pre-hardened steel moulds are less wear resistant and used for production lines where overall volume productions are low. The moulds have two basic components:
  - The injection mould (A plate)
  - The ejector mould (B plate)
- **Injection of molten plastic:** A precise amount of molten plastic, called a "*shot*" in industry parlance is injected through the sprue into the mould. The molten plastic covers the cavity within and takes the shape of the negative space within the mould. A continuous pathway through the moulds is provided to allow the coolant (usually water) to flow and facilitate an exchange of heat from the mould which has absorbed heat from the molten plastic. This keeps the mould at an optimum temperature.
- **Ejection of product:** The formed plastic product is ejected from the mould by the cylindrical ejector pins placed in the ejector mould.

5.2. General Quality Checks

Following are a few visual quality parameters/checks:

- Check for production defects, such as blistering, burn marks, air voids etc.
- Check for uniformity of colour (Check Pantone colour chart)
- Verify that individual parts adhere to measurement specifications
- Check for any product uniformity issues, including appearance, volume, weight and geometric measurement

6. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

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- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

6.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	**a.** Plastic Manufacture, Sale and Usage Rules (2009) **b.** Food Safety and Standards (Packaging) Regulations, 2018	**a.** CPCB/SPCB (Central/State Pollution Control Boards) **b.** Food Safety and Standards Authority of India (FSSAI)	**a.** Prohibition on use of recycled plastics for storing, carrying, dispensing or packaging of food items. **b.** General and specific requirements for food packaging materials

6.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	European Union	**a.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations (EC) 1907/2006 **b.** Regulation (EC) No. 1935/2004 **c.** Regulation (EC) No. 2023/2006 on GMP **d.** Regulation (EU) No. 10/2011	**a.** European Commission **b.** European Chemicals Agency (ECHA)	**a.** Hazardous chemicals **b.** Persistent organic pollutants **c.** General product safety directive **d.** Food contact materials (FCM) labelling requirements, traceability and declaration of compliance by manufacturers. **e.** Requirements for manufacturing and marketing of Food contact materials (FCM), Plastics
2	United States of America	**a.** Consumer Product Safety Commission(CPSC), Act of 2008 **b.** Consumer Product Safety Act (CPSA), Act of 1972 **c.** Federal Hazardous Substances Act (FHSA), Act of 1960.	**a.** Consumer Product Safety Commission (CPSC) **b.** California State	**a.** Regulations on chemicals **b.** Quantity of regulated material **c.** Product labelling **d.** Surface flammability

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
		d. California Proposition 65, Act of 1986		
3	Australia	a. Recycling and Waste Reduction, Act 2020 b. AS 2070: 1999 Regulations	a. Department of Agriculture, Water and Environment	a. Regulation of exports b. Encouraging and regulating manufacturers, importers, distributors, designers and other persons to take responsibility for products, including by taking action that relates to: c. Reducing or avoiding generating waste through improvements in product design d. Improving the durability, reparability and reusability of products e. Managing products throughout their life cycle f. Regulation of Food contact plastics
4	Middle East	a. GSO Technical Regulations: i. GSO 2231: 2012 ii. GSO 839/1997	a. GCC Standardization Organization (GSO)	a. General Requirements For The Materials Intended To Come Into Contact With Food

7. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 2828: 2019**	Details about terms used in plastic industry	Middle East	**GSO ISO 21067: 2010: 2010**	Vocabulary of packaging and material handling
	IS 7019: 1998 (Reaffirmed Year: 2018)	Details about terms used in plastic packaging industry	Australia	**AS 2609**	Methods for assessing modifications to the flavour of foodstuffs due to packaging, Method 1: Sensory analysis
	IS 7792: 1975 (Reaffirmed Year: 2018)	Code regarding the practice for handling plastic containers.	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	**IS 3395: 1997 (Reaffirmed Year: 2020)**	Details about Low Density Polyethylene (LDPE)	Australia	**ISO 24022-2: 2020**	Plastics — Polystyrene (PS) moulding and extrusion materials — Part 2: Preparation of test specimens and determination of properties
	IS 2267: 1995 (Reaffirmed Year: 2020)	Polystyrene Moulding and Extrusion Materials	USA	**ASTM D4549**	Standard Classification System and Basis for

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
					Specification for Polystyrene and Rubber-Modified Polystyrene Moulding and Extrusion Materials (PS)
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 10106: Part 1: Sec 1: 1990 (Reaffirmed Year: 2021)**	Guidelines on the packing foodstuffs	European Union	**ISO 293: 2004**	Compression moulded test specimens of thermoplastics materials
	IS 10106: Part 2: Sec 3: 1984 (Reaffirmed Year: 2019)	Description of the properties and characteristics of plastic materials	European Union	**ISO 178: 1975**	Flexural properties of rigid plastics
	IS 2530: 1963 (Reaffirmed Year: 2018)	Methods of test of moulding materials	-	-	-
	IS 9833: 2018	List of colourants for use in plastics in contact with foodstuffs and pharmaceuticals (second revision)	-	-	-
	IS 2798: 1998 (Reaffirmed Year: 2018)	Methods of test for plastics containers	-	-	-
Finishing (Refer Table 4.a & 4.b)	**IS 13360: Part 1 : 1992 (Reaffirmed Year: 2018)**	Methods for testing	European Union	**ISO 1183 - 3: 1999**	Density of solid plastics
	IS 196:1966 (Reaffirmed Year: 2022)	Atmospheric conditions for testing plastic	USA	**ASTM D2583**	Hardness of plastics
	-	-	USA	**ASTM D3835**	Rheological properties of thermoplastics

7.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 2828: 2019** - Plastics - Vocabulary	▪ Define the terms used in plastic industry, English and French ▪ Synonymous terms follow the preferred term
	IS 7019: 1998 (Reaffirmed Year: 2018) - Glossary of terms in plastics and flexible packaging, excluding paper	▪ Defines the packing terms used in plastics and flexible packaging excluding paper
	IS 7792: 1975 (Reaffirmed Year: 2018) - Code of practice for handling plastic containers	▪ Covers broadly the safe and proper transportation, handing and storage ▪ Aspects of the plastic container both filled and empty
	GSO ISO 21067: 2010 - Packaging - Vocabulary	▪ Specifies preferred terms and definitions related to packaging and materials handling, for use in international commerce

Stage	Code	Scope
	AS 2609 - Methods for assessing modifications to the flavour of foodstuffs due to packaging, Method 1: Sensory analysis	▪ Sensory analysis of food and impact of packaging materials

7.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 3395: 1997 (Reaffirmed Year: 2020)** - Low density polyethylene(LDPE) And linear low density polyethylene(LLDPE) materials for moulding and extrusion- specification	▪ Intended to be used for identification and characterization of LDPE and linear low density polyethylene ▪ Indicates the major end use(s), it does not impose any restriction ▪ Does not cover master batches
Raw Material	**IS 2267: 1995 (Reaffirmed Year: 2020)** - Polystyrene moulding and extrusion materials - Specification	▪ Requirements of polystyrene ▪ Tests ▪ Normative references and adjunct standards

7.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**ISO 24022- 2: 2020** - Plastics — Polystyrene (PS) moulding and extrusion materials — Part 2: Preparation of test specimens and determination of properties	▪ Test methodology ▪ Preparation of test specimens
Raw Material	**ASTM D4549** - Standard Classification System and Basis for Specification for Polystyrene and Rubber-Modified Polystyrene Moulding and Extrusion Materials (PS)	▪ Polystyrene materials: Crystal and rubber modified for extrusion and moulding requirements ▪ Exclusions ▪ Mention of SI units

7.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 10106: Part 1: Sec 1: 1990 (Reaffirmed Year: 2021) -** Packaging code, Part 1: Product packaging, section 1 foodstuffs and perishables	▪ Guidelines on the packaging of foodstuffs and perishables ▪ Type of material for packaging for food material
Processing	**IS 10106: Part 2: Sec 3: 1984 (Reaffirmed Year: 2019) -** Packaging code, Part 2: Packaging materials, Section 3: Plastics materials	▪ Description of the properties and characteristics of plastic materials and guidelines for the choice of plastics suitable for different uses
Processing	**IS 2530: 1963 (Reaffirmed Year: 2018)** - Methods of test for polyethylene moulding materials and polyethylene compounds	▪ Prescribes the methods of test for polyethylene moulding materials and polyethylene compounds ▪ Some are for apparent density, bulk factor, and elongation at break and melt flow
Processing	**IS 2798: 1998 (Reaffirmed Year: 2018)** - Methods of test for plastics containers	▪ Prescribes the methods of test for plastics containers ▪ Measurement test for some aspects overall height, diameter , neck height etc.
Processing	**IS 9833: 2018** - List of colourants for use in plastics in contact with foodstuffs and pharmaceuticals (second revision)	▪ Lists of permitted colourants ▪ Referenced documents ▪ Test methodology

Stage	Code	Scope
	IS 2798: 1998 (Reaffirmed Year: 2018): Methods of test for plastics containers	▪ Method of test for compatibility of plastic container

7.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**ISO 178: 1975** - Plastics — Determination of flexural properties	▪ A method for determining the flexural properties of rigid and semi-rigid plastics under defined conditions ▪ Thermoplastic moulding, extrusion and casting materials, including filled and reinforced compounds in addition to unfilled types
	ISO 293: 2004 - Plastics — Compression moulding of test specimens of thermoplastic materials	▪ Specifies the general principles and the procedures to be followed with thermoplastics in the preparation of compression-moulded test specimens ▪ The main steps of the procedure, including four different cooling methods, are standardized. ▪ The required moulding temperature and cooling methods are as specified in the appropriate International Standard for the material or as agreed between the interested parties

7.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 13360: Part 1 : 1992 (Reaffirmed Year: 2018)** - Plastics - Methods of Testing, Part 1: Introduction	▪ Methods for testing. It covers standard atmospheres for conditioning and testing plastics
	IS 196:1966 (Reaffirmed Year: 2022) - Atmospheric conditions for testing	▪ Atmospheric conditions for testing plastic ▪ Specifies the atmospheric conditions for testing of materials, products, equipment, etc.

7.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**ISO 1183 - 3: 1999** - Plastics — Methods for determining the density and relative density of non-cellular plastics	▪ Specification of four methods for the determination of the density of plastics except cellular plastics ▪ Immersion method for plastics in a finished condition ▪ Pyknometer method for plastics in the form of small particles such as powders, granules pellets or flakes
	ASTM D2583 - 13a: Standard test method for indentation hardness of rigid plastics by means of a barcol impressor	▪ Method covers the determination of indentation hardness of both reinforced and non-reinforced rigid plastics
	ASTM D3835: Standard test method for determination of properties of polymeric materials by means of a capillary rheometer	▪ Covers measurement of the rheological properties of polymeric materials at various temperatures and shear rates common to processing equipment ▪ measurement of melt viscosity, sensitivity, or stability of melt viscosity with respect to temperature and polymer

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

1. Product information- http://odopup.in/en/article/kanpur-dehat
2. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
3. ISO Standards: https://www.iso.org/obp/ui#search
4. EU: https://www.en-standard.eu/
5. USA: https://www.en-standard.eu/astm-standards/ , https://www.astm.org/
6. AUS: https://www.standards.org.au/
7. GULF: https://www.gso.org.sa/store/
8. NABL accredited laboratories: https://nabl-india.org/ , https://parakh.ncog.gov.in
9. Domestic Legislation/Regulation: https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
10. International Legislation/Regulation:
    a. EU: https://www.compliancegate.com/plastic-product-regulations-european-union/
    b. USA: https://www.compliancegate.com/single-use-plastic-regulations-united-states/
    c. AUS: https://www.abcb.gov.au/resources/filter/standards-and-protocols ; https://www.compliancegate.com/food-contact-material-regulations-australia/
    d. GULF (GCC): https://www.saso.gov.sa/Documents/SASO-Technical-Regulations-English.pdf

PLASTIC CHAIRS, KANPUR DEHAT

1. INTRODUCTION

The chair has been used since antiquity, although for many centuries it was a symbolic article of state and dignity rather than an article for ordinary use. Chairs are amongst the most basic piece of furniture in the Indian household. It serves to be major furniture for sitting and relaxing. Kanpur Dehat is known for production of plastic products. There are various types of plastic chairs which are manufactured here.

Its primary features are two pieces of a durable material, attached as back and seat to one another at a 90° or slightly greater angle, with usually the four corners of the horizontal seat attached in turn to four legs—strong enough to support the weight of a person who sits on the seat (usually wide and broad enough to hold the lower body from the buttocks almost to the knees) and leans against the vertical back (usually high and wide enough to support the back to the shoulder blades). Used in a number of rooms in homes, in schools and offices (with desks), and in various other workplaces, The monobloc chair is a lightweight stackable polypropylene chair, usually white in colour, often described as the world's most common plastic chair.

2. RAW MATERIALS

To make a plastic chair, the raw materials used are:

- Polyethylene terephthalate (PET): PET has high mechanical strength as well as sufficient stiffness and rigidity. When we consider the chair as a product, strength is an important criterion. Hence PET is used as a raw material to produce the monobloc plastic chair. PET is resistant against solvents which are a major asset as products made out of PET can be easily cleaned by using water as well as chemical solvents.
- Polypropylene
- Any thermoplastic which can be recycled

3. MANUFACTURING PROCESS

The manufacturing of plastic furniture is an injection moulding process. Injection moulding is process of manufacturing plastic products at a very rapid rate. This is a cost intensive process and only feasible where large production volumes offset the steep initial costs for manufacturing of the moulds and machinery capex investment. Various steps involved in the injection moulding process are as follows:

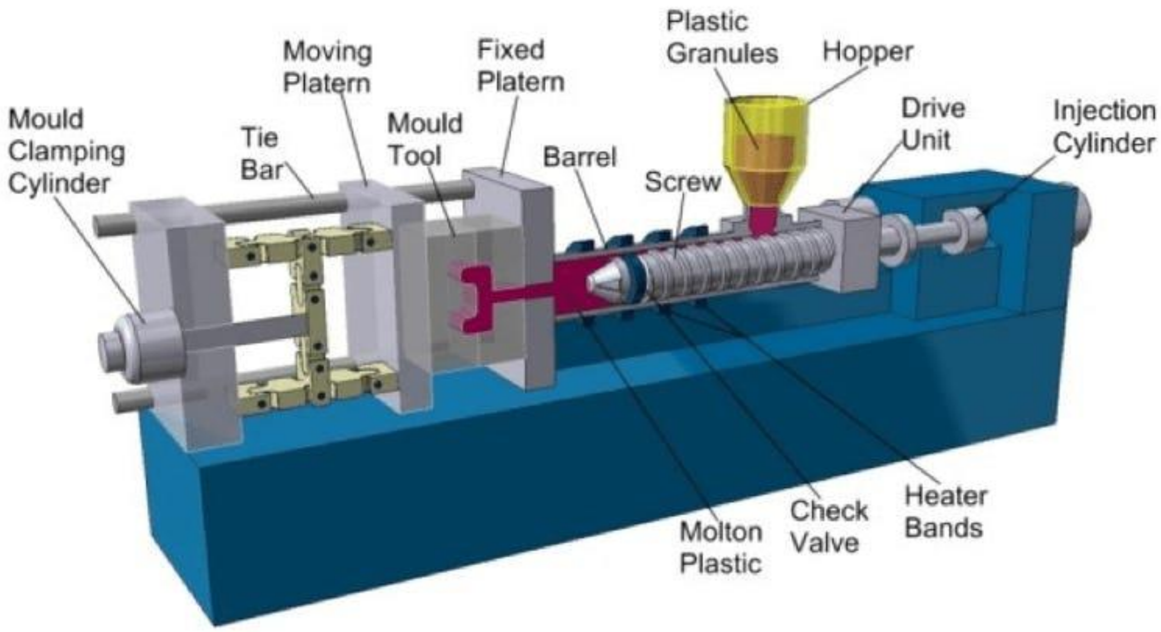

Figure 1: Typical Flow Process for Injection Moulding

3.1. Mould and mould design

The choice of materials to manufacture the moulds depends largely on the economics of the operation. Steel moulds are the most expensive to make but their longer life span offsets the high initial cost while pre-hardened steel moulds are less wear resistant and used for production lines where overall volume productions are low. The moulds have two basic components:

- The injection mould (A plate)
- The ejector mould (B plate)

3.2. Injection of molten plastic

A precise amount of molten plastic, called a "*shot"* in industry parlance is injected through the sprue into the mould. The molten plastic covers the cavity within and takes the shape of the negative space within the mould.

A continuous pathway through the moulds is provided to allow the coolant (usually water) to flow and facilitate an exchange of heat from the mould which has absorbed heat from the molten plastic. This keeps the mould at an optimum temperature.

3.3. Ejection of product

The formed plastic product is ejected from the mould by the cylindrical ejector pins placed in the ejector mould.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for production defects, such as blistering, burn marks, air voids etc.
- Check for uniformity of colour (Check Pantone colour chart)

- Verify that individual parts adhere to measurement specifications
- Check for any product uniformity issues, including appearance, volume, weight and geometric measurement

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

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5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
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5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
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2	United States of America	a. Consumer Product Safety Improvement Act (CPSIA), Act of 2008 b. Consumer Product Safety Act (CPSA) Act of 1972 c. Federal Hazardous Substances Act (FHSA), Act of 1960. d. California Proposition 65, Act of 1986	a. Consumer Product Safety Commission (CPSC) b. California State	a. Regulations on chemicals b. Quantity of regulated material c. Product labelling d. Surface flammability
3	Australia	a. Chemicals and Heavy Metals Regulations	a. Australia Building	a. Product safety rules and standards

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
			Codes Board (ABCB)	
4	Middle East	**a.** Saudi Standards, Metrology and Quality Organization (SASO) Regulations **b.** SASO Certificate of Conformity	**a.** Saudi Standards, Metrology and Quality Organization (SASO)	**a.** General product safety regulation **b.** Labelling requirement

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 2828: 2019**	Details about terms used in plastic industry	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	**IS 13713: 2021**	Specifications about materials, dimensions, methods of test and acceptance criteria for chairs made of plastic materials	European Union	**ISO 293: 2004**	Compression moulded test specimens of thermoplastics materials
	IS 2530: 1963 (Reaffirmed Year: 2018)	Methods of test of moulding materials	European Union	**ISO 1183 - 3: 1999**	Density of solid plastics
	-	-	European Union	**ISO 178: 1975**	Flexural properties of rigid plastics
	-	-	USA	**ASTM D3835**	Rheological properties of thermoplastics
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 5416: Part 1: 1998 (Reaffirmed Year: 2021) IS 5416: Part 2: 1998 (Reaffirmed Year: 2021)**	Tests for determining the stability and strength of folding chairs.	USA	**ASTM D2583**	Hardness of plastics
	IS 3663: 2018	Specifications about the dimensions of the chairs	USA	**ASTM F1561-03**	Requirements and standard for plastic chairs for outdoor use
	IS 13360: Part 1 : 1992 (Reaffirmed Year: 2018)	Plastics - Methods of testing, Part 1: Introduction	USA	**ASTM F1838**	Performance requirements for adult and children's plastic chairs for outdoor use
	IS 196:1966 (Reaffirmed Year: 2022)	Atmospheric conditions for testing plastic	Australia	**AS/NZS 3813: 2016**	Determination of strength and durability, stability, UV and weathering, and ignitability
	-	-	Gulf Countries	**GSO ISO 7174-1: 2016**	Stability assessment

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
Finishing (Refer Table 4.a & 4.b)	**IS 13360: Part 1: 1992 (Reaffirmed Year: 2018)**	Plastics - Methods of Testing	Gulf Countries	**GSO 1561**	Performance requirements for Plastic Chairs For Outdoor use
	-	-	Gulf Countries	**GSO 1593: 2002**	Methods of testing chairs

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 2828: 2019** - Details about terms used in plastic industry	▪ Details about terms used in plastic industry

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 13713: 2021** - Specifications about materials, dimensions, methods of test and acceptance criteria for chairs made of plastic materials.	▪ This standard specifies materials, dimensions, methods of test and acceptance criteria for chairs made of plastic materials in combination with steel tubes or as one-piece plastic moulded chairs.
	IS 2530: 1963 (Reaffirmed Year: 2018) - Methods of test of moulding materials	▪ This standard prescribes the methods of test for polyethylene moulding materials and polyethylene compounds.

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**ISO 293: 2004** - Compression moulded test specimens of thermoplastics materials	▪ This International Standard specifies the general principles and the procedures to be followed with thermoplastics in the preparation of compression-moulded test specimens, and sheets from which test specimens may be machined or stamped ▪ In order to obtain mouldings in a reproducible state, the main steps of the procedure, including four different cooling methods, are standardized. For each material, the required moulding temperature and cooling methods are as specified in the appropriate International Standard for the material or as agreed between the interested parties
	ISO 1183 - 3: 1999 - Density of solid plastics	▪ This document specifies three methods for the determination of the density of non-cellular plastics in the form of void-free moulded or extruded objects, as well as powders, flakes and granules — Method A: Immersion method, for solid plastics (except for powders) in void-free form — Method B: Liquid pycnometer method, for particles, powders, flakes, granules or small pieces of finished parts — Method C: Titration method, for plastics in any void-free form

Stage	Code	Scope
	ISO 178: 1975 - Flexural properties of rigid plastics	▪ This document specifies a method for determining the flexural properties of rigid and semi-rigid plastics under defined conditions. A preferred test specimen is defined, but parameters are included for alternative specimen sizes for use where appropriate. A range of test speeds is included ▪ The method is used to investigate the flexural behaviour of the test specimens and to determine the flexural strength, flexural modulus and other aspects of the flexural stress/strain relationship under the conditions defined. It applies to a freely supported beam, loaded at mid-span (three-point loading test)
	ASTM D3835 - Rheological properties of thermoplastics	▪ This test method covers measurement of the rheological properties of polymeric materials at various temperatures and shear rates common to processing equipment. It covers measurement of melt viscosity, sensitivity, or stability of melt viscosity with respect to temperature and polymer dwell time in the rheometer, die swell ratio (polymer memory), and shear sensitivity when extruding under constant rate or stress. The techniques described permit the characterization of materials that exhibit both stable and unstable melt viscosity properties. ▪ This test method has been found useful for quality control tests on both reinforced and unreinforced thermoplastics, cure cycles of thermosetting materials, and other polymeric materials having a wide range of melt viscosities

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 5416: Part 1: 1998 (Reaffirmed Year: 2021)** - Part 1 Strength **IS 5416: Part 2: 1998 (Reaffirmed Year: 2021): Part 2** - Determination of stability of chairs and stools	▪ This standard (Part 1) gives test methods for determination of strength of the structure of all types of chairs and stools ▪ This standard (Part 2) prescribes the methods for determination of stability of all types of upright chairs and stools, easy chairs and tilting and reclining chairs
	IS 3663: 2018 - Specifications about the dimensions of the chairs	▪ This standard specifies essential dimensions of tables and chairs for office purposes
	IS 13360: Part 1 : 1992 (Reaffirmed Year: 2018) - Plastics - Methods of testing, Part 1: Introduction	▪ This Indian Standard gives a general introduction to IS 13360: 1992, which describes a range of methods for testing-plastics ▪ It covers standard atmospheres for conditioning and testing plastics. The individual methods published/ being prepared are listed in Annex A

Stage	Code	Scope
	IS 196:1966 (Reaffirmed Year: 2022) - Atmospheric conditions for testing plastic	▪ This standard specifies the atmospheric conditions for testing of materials, products, equipment, etc., and applies to such tests where atmospheric conditions need to be controlled to obtain comparable and reproducible results or to conduct measurements where test results obtained under different conditions have to be reduced to standard conditions ▪ This standard does not apply to the basic standards of weights and measures, to the precision measurement made in terms of these basic standards and to such tests as calibration of test gauges, precision tools, etc., and to the cases covered in 0.9

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**ASTM D2583** - Hardness of plastics	▪ This test method covers the determination of indentation hardness of both reinforced and no reinforced rigid plastics using a Barcol Impressor, Model No. 934-1 and Model No. 935 ▪ The values stated in SI units are to be regarded as standard ▪ The values given in parentheses are for information only ▪ This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use
	ASTM F1561-03 - Requirements and standard for plastic chairs for outdoor use.	▪ These requirements establish nationally recognized performance requirements for Class A (residential) and Class B (non-residential) plastic chairs intended for outdoor use ▪ These requirements do not address conditions that affect the performance of the chair beyond the manufacturing site ▪ These requirements are not applicable to chaises, multi-positional chairs, upholstered chairs, or other types of furniture
	ASTM F1838 - Performance requirements for adult and children's plastic chairs for outdoor use.	▪ These standard performance requirements establish nationally recognized performance requirements for Class A (residential) and Class B (non-residential) adult and children's plastic chairs intended for outdoor use ▪ These requirements do not address conditions that affect the performance of the chair beyond the manufacturing site ▪ These standard performance requirements are not applicable to chaises, multi-positional chairs, upholstered chairs, or other types of furniture
	AS/NZS 3813 - Determination of strength and durability, stability, UV and weathering, and ignitability	▪ This Standard sets out requirements for the evaluation and selection of plastic monobloc chairs for adults but does not include chairs intended for bathroom use. It specifies minimum requirements for strength, durability and stability of the completed chair, but does not account for materials, design, construction or the process of manufacture

Stage	Code	Scope
	GSO ISO 7174-1: 2016 - Stability assessment	▪ This part of ISO 7174 describes methods for determining the stability of all types of upright chairs, stools and pouffes. It does not apply to settees and other multiple seating, nor to reclining chairs when they are reclined, chairs with tilting mechanisms when they are tilted, nor to swivelling or rocking chairs. The methods are, however, applicable to testing chairs with reclining, tilting and adjustable back-angle mechanisms when these are used as upright chairs ▪ Part 2 of ISO 7174 deals with stability for chairs with tilting or reclining mechanisms when fully reclined. The test results are only valid for the article tested. When the test results are intended to be applied to other similar articles, the test specimen should be representative of the production model. In the case of designs not catered for in the test procedures, the test should be carried out as far as possible as described, and deviations from the test procedure recorded in the test report

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 13360: Part 6: Section 6: 1999 (Reaffirmed Year: 2022)** - Methods of testing plastic (Part6/Sec 6)	▪ Analysis of thermal properties of plastic chair
	IS 13360: Part 8: Section 13: 2021 - Methods of testing plastic (Part8/Sec 13)	▪ Analysis of chemical properties and performance of plastic chair ▪ Determination of changes in colour and variations in properties after exposure to daylight under glass, natural weathering or laboratory light sources
	IS 13360: Part 11: Section 14: 2006 (Reaffirmed Year: 2021) - Methods of testing plastic (Part11/Sec 14)	▪ Determination of dimensional change on heating

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**GSO 1561** - Performance requirements for plastic chairs for outdoor use	▪ These requirements establish nationally recognized performance requirements for Class A (residential) and Class B (non-residential) plastic chairs intended for outdoor use. ▪ These requirements do not address conditions that affect the performance of the chair beyond the manufacturing site
	GSO 1593: 2002 - Methods of testing chairs	▪ Methods of testing chairs

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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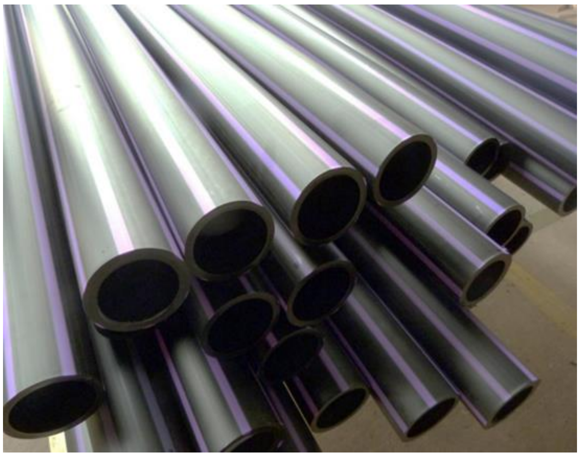
3. ISO Standards: https://www.iso.org/obp/ui#search
4. EU: https://www.en-standard.eu/
5. USA: https://www.en-standard.eu/astm-standards/, https://www.astm.org/
6. AUS: https://www.standards.org.au/
7. GULF: https://www.gso.org.sa/store/
8. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in
9. Domestic Legislation/Regulation: https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
10. International Legislation/Regulation:
    a. EU: https://ec.europa.eu/growth/sectors/chemicals/reach_en
    b. USA: https://www.cpsc.gov/Regulations-Laws-Standards/Statutes/Summary-List/Consumer-Product-Safety-Improvement-Act-CPSIA; https://www.cpsc.gov/Business-Manufacturing/Business-Education/Business-Guidance/FHSA-Requirements
    c. AUS: https://www.abcb.gov.au/
    d. GULF (GCC): https://www.saso.gov.sa/Documents/SASO-Technical-Regulations-English.pdf

PLASTIC PIPES, KANPUR DEHAT

1. INTRODUCTION

A Plastic pipe is a tubular section, or hollow cylinder, made of plastic. It is usually, but not necessarily, of circular cross-section, used mainly to convey substances which can flow—liquids and gases (fluids), slurries, powders and masses of small solids. It can also be used for structural applications; hollow pipes are far stiffer per unit weight than solid members. Plastic pipe work is also used for the conveyance of potable water, waste water, heating fluids and cooling fluids, foodstuffs, ultra-pure liquids, slurries, gases, compressed air, irrigation, plastic pressure pipe systems, and vacuum system applications amongst several other uses. Kanpur Dehat is well known for production of plastic products including plastic pipes.



There are various types of plastic pipes which are manufactured and are used as per the requirement. There are three basic types of plastic pipes:

- **Solid wall pipe:** Pipes consisting of one layer of a homogeneous matrix of thermoplastic material which is ready for use in a pipeline.
- **Structured wall pipe:** Structured-wall pipes and fittings are products which have an optimized design with regard to material usage to achieve the physical, mechanical and performance requirements. Structured wall pipes are tailor made solutions of piping systems, for a variety of applications and in most cases developed in cooperation with users.
- **Barrier pipe:** Pipes incorporating a flexible metallic layer as the middle of three bonded layers. Barrier pipe is used, for example, to provide additional protection for the contents passing through the pipe (particularly drinking water) from aggressive chemicals or other pollution when laid in ground contaminated by previous use.

Most plastic pipe systems are made from thermoplastic materials. The production method involves melting the material, shaping and then cooling. Pipes are produced by extrusion. Plastic pipe systems fulfil a variety of service requirements such as:

- Conveyance of gas: Highest Safety requirements
- Plastic pipes for radiant heating and floor heating:
- Sewer applications: High chemical resistance

2. RAW MATERIALS

To make plastic pipes, generally the raw materials used are:

- **Chlorinated PVC:** It is resistant to many acids, bases, salts, paraffinic hydrocarbons, halogens and alcohols although it is not resistant to solvents, aromatics and some chlorinated hydrocarbons. It can carry higher temperature liquids than UPVC with a max operating temperature reaching 200 °F (93.3 °C). Due to its greater temperature threshold and chemical resistance, CPVC is one of the main recommended material choices in residential, commercial, and industrial water and liquid transport.
- Any thermoplastic (a plastic polymer material that becomes pliable or moldable at a certain elevated temperature and solidifies upon cooling)

3. Manufacturing Process

PVC pipes are commonly used for manufacturing sewage pipes, water mains and irrigation pipes. Possessing long-lasting properties, PVC pipes are easy to install, lightweight, strong, durable and easily recyclable, making them cost-efficient and sustainable. The smooth surface of PVC pipes also encourages faster water flow due to lower amounts of friction than piping made from other materials such as cast iron or concrete. PVC pipes can also be manufactured to varying lengths, to suit different wall thicknesses and diameters.

PVC pipes are manufactured by extrusion of PVC; a brief process flow can be seen as below:

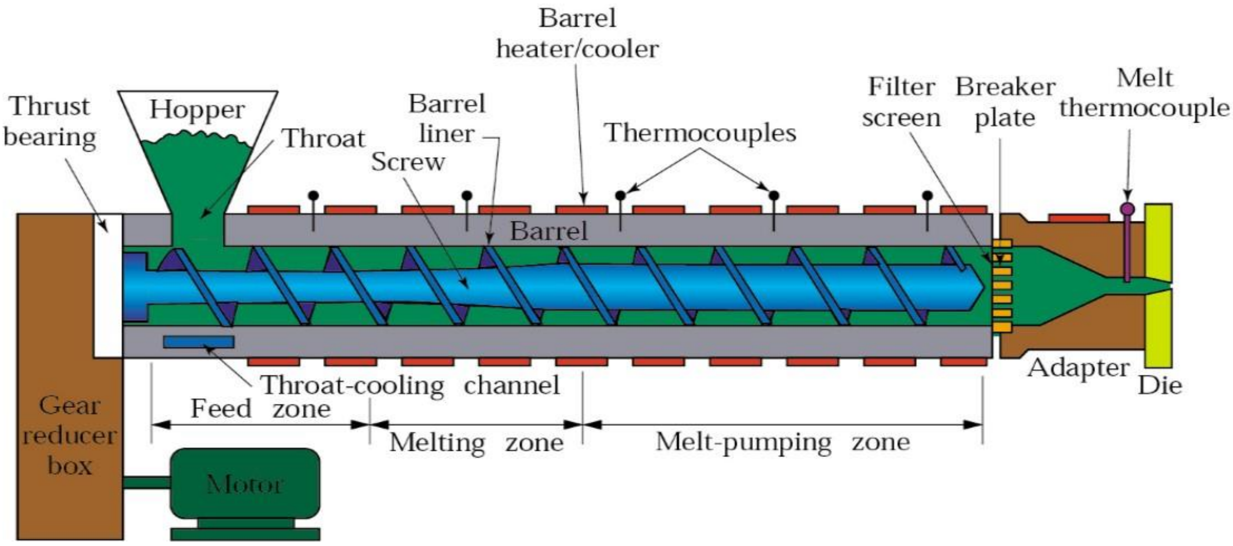


Figure 1: A typical Pipe Extrusion process flow

The PVC pellets with other additives based on the specifications required, are fed into the feed hopper. Material is fed from the hopper into the barrel of the screw extruder. The screw spreads the mixture through the length of the barrel. The heaters along the barrel increase the temperature up to 275 °C, melting the mixture of PVC pellets and additives. The melted mixture is forced into a shaping die through a feeder. The extruded pipe sections are immersed in a sealed water bath to accelerate the process of cooling. Upon cooling, the pipes are cut to required sizes and packed.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for any product uniformity issues, including appearance, volume, weight and geometric measurement.
- Check for production defects, such as bumps, depressions, chipping and cracks.
- The colour of the pipe should be uniform
- Check the weight of the assembled product and individual parts for compliance with specifications.

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	a. Lead Stabilizer in Polyvinyl Chloride (PVC) Pipes and Fittings Rules, 2021	a. CPCB/SPCB (Central/State Pollution control Board)	a. Prohibition of lead as a stabilizer in PVC pipes and fittings

5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	European Union	a. Registration, Evaluation, Authorization and Restriction of	a. European Commission	a. Hazardous chemicals b. Persistent organic pollutants

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
		Chemicals (REACH) Regulations (EC) 1907/2006	**b.** European Chemical Agency (ECHA)	**c.** General product safety directive
2	United States of America	**a.** Consumer Product Safety Improvement Act (CPSIA), Act of 2008 **b.** Consumer Product Safety Act (CPSA) Act of 1972 **c.** Federal Hazardous Substances Act (FHSA), Act of 1960. **d.** California Proposition 65, Act of 1986	**a.** Consumer Product Safety Commission (CPSC) **b.** California State	**a.** Regulations on chemicals **b.** Quantity of regulated material **c.** Product labelling **d.** Surface flammability
3	Australia	**a.** Chemicals and Heavy Metals Regulations	**a.** Australia Building Codes Board (ABCB)	**a.** Product safety rules and standards
4	Middle East	**a.** Saudi Standards, Metrology and Quality Organization (SASO) Regulations **b.** SASO Certificate of Conformity	**a.** Saudi Standards, Metrology and Quality Organization (SASO)	**a.** General product safety regulation **b.** Labelling requirement

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 2828: 2019**	Details about terms used in plastic industry	-	-	-
	IS 7019: 1998 (Reaffirmed Year: 2018)	Details about terms used in plastic packaging industry	-	-	-
	IS 10151: 2019	Safety use of PVC and its copolymers in contact with foodstuffs, pharmaceuticals and drinking water	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	**IS 4905: 2015 (Reaffirmed Year: 2020)**	Methods of random sampling	European Union	**ISO 293: 2004**	Compression moulded test specimens of thermoplastics materials
	IS 7634: Part 1: 1975 (Reaffirmed Year: 2017)	Selection of plastic pipe systems for cold water services up to 37°C and general recommendations applicable to all types of plastic pipe systems	European Union	**ISO 1183 - 3: 1999**	Density of solid plastics
	-	-	European Union	**ISO 178: 1975**	Flexural properties of rigid plastics

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
	-	-	European Union	**ISO R/527**	Tensile properties of plastics
	-	-	Gulf Countries	**GSO ISO 265-1:2013**	Fittings for domestic and industrial waste pipes - Basic
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 4985: 2021**	Requirements for plain as well as socket-ended pipes, including those for use with elastomeric sealing rings, for potable water supplies	European Union	**ISO 3127: 1994**	Determination of resistance to external blows
	IS 12235: Part 1 TO 19: 2004 (Reaffirmed Year: 2019)	Thermoplastic's – Pipes and fittings	USA	**ASTM D3835**	Rheological properties of thermoplastics
	IS 196:1966 (Reaffirmed Year: 2022)	Atmospheric conditions for testing	Australia	**AS/NZS 4765: 2017**	Specifications of PVC pipes for pressure applications
	IS 2530: 1963 (Reaffirmed Year: 2018)	Methods of test of moulding materials	Gulf Countries	**GSO ISO 6259-2: 2015**	Determination of tensile properties
	IS 9271: 2004 (Reaffirmed Year: 2014)	Specification and requirement of UPVC pipes used for drainage.	-	-	-
	IS 12818: 2010 (Reaffirmed Year: 2021)	Specification and requirement of UPVC pipes used for bore/tube well	-	-	-
	IS 14787: 2000 (Reaffirmed Year: 2020)	Specification and requirement of UPVC pipes used for underground cabling	-	-	-
Finishing (Refer Table 4.a & 4.b)	**IS 13360: Part 1 : 1992 (Reaffirmed Year: 2018)**	Methods for testing. It covers standard atmospheres for conditioning and testing plastics.	USA	**ASTM D2583**	Hardness of plastics
	IS 4669: 1968 (Reaffirmed Year: 2018)	Methods to test PVS resins	USA	**ASTM F441**	Sizing standard- length, thickness and diameter
	-	-	Australia	**AS/NZS 1462.19:2006**	Methods of test for plastic pipes and fittings
	-	-	Gulf Countries	**GSO ISO 4439: 2015**	Determination and specification of density

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 2828: 2019** - Plastics - Vocabulary	▪ Define the terms used in plastic industry, (English and French) ▪ Synonymous terms follow the preferred term
	IS 7019: 1998 (Reaffirmed Year: 2018) - Glossary of terms in plastics and flexible packaging, excluding paper	▪ Defines the packing terms used in plastics and flexible packaging excluding paper
	IS 10151: 2019 - Polyvinyl Chloride (PVC) and its co-polymers for its safe use in contact with foodstuffs, pharmaceuticals and drinking water	▪ Safety use of PVC and its copolymers in contact with foodstuffs, pharmaceuticals and drinking water ▪ Applicable to resin or finished articles

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 4905: 2015 (Reaffirmed Year: 2020)** - Methods of random sampling	▪ Procedures for the selection of items from a lot on an objective basis by using random sampling techniques ▪ Describes the methods of calculation of simple estimates
	IS 7634: Part 1: 1975 (Reaffirmed Year: 2017) - Code of practice for plastic pipes work for potable water supplies	▪ Code of practice for plastic pipes work for potable water supplies ▪ Choice of materials and general recommendation

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**ISO 293: 2004** - Plastics — Compression moulding of test specimens of thermoplastic materials	▪ Specifies the general principles and the procedures to be followed with thermoplastics in the preparation of compression-moulded test specimens ▪ The main steps of the procedure, including four different cooling methods, are standardized. ▪ The required moulding temperature and cooling methods are as specified in the appropriate International Standard for the material or as agreed between the interested parties.
	ISO 178: 1975 - Plastics — Determination of flexural properties	▪ A method for determining the flexural properties of rigid and semi-rigid plastics under defined conditions ▪ Thermoplastic moulding, extrusion and casting materials, including filled and reinforced compounds in addition to unfilled types
	ISO 1183 - 3: 1999 - Plastics — Methods for determining the density and relative density of non-cellular plastics	▪ Specification of four methods for the determination of the density of plastics except cellular plastics ▪ Immersion method for plastics in a finished condition ▪ Pyknometer method for plastics in the form of small particles such as powders, granules pellets or flakes

Stage	Code	Scope
	ISO 178: 1975 - Plastics — Determination of flexural properties	▪ A method for determining the flexural properties of rigid and semi-rigid plastics under defined conditions ▪ Thermoplastic moulding, extrusion and casting materials, including filled and reinforced compounds in addition to unfilled types
	GSO ISO 265-1:2013 - Pipes and fittings of plastics materials - Fittings for domestic and industrial waste pipes - Basic dimensions: Metric series - Part 1: Un-plasticized polyvinyl chloride (PVC-U)	▪ Specifies the series of diameters and the formulae for calculation of the dimensions common to the main types of fittings with spigot ends (socket fittings with curved entries and branches) made of PVC-U

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 4985: 2021** - Plasticized PVC pipes for potable water supplies	▪ Covers requirement for plain as well as socket-ended pipes ▪ Including those for use with elastomeric sealing rings , for potable water supplies
	IS 12235: Part 1 TO 19: 2004 (Reaffirmed Year: 2019) - Thermoplastics pipes and fittings-methods of test	▪ specifies the method for measurement of outside diameter, wall thickness, ▪ length, and internal diameters and depths of pipe sockets of thermoplastics pipes and fittings,
	IS 196:1966 (Reaffirmed Year: 2022) - Atmospheric conditions for testing	▪ Atmospheric conditions for testing plastic ▪ Specifies the atmospheric conditions for testing of materials, products, equipment, etc.,
	IS 2530: 1963 (Reaffirmed Year: 2018) - Methods of test for polyethylene moulding materials and polyethylene compounds	▪ This standard prescribes the methods of test for polyethylene moulding materials and polyethylene compounds.
	IS 9271: 2004 (Reaffirmed Year: 2014) - Plasticized polyvinyl chloride (UPVC) single wall corrugated pipes for drainage - Specification	▪ Specifies the requirements and methods of test of un-plasticized PVC corrugated perforated/non-perforated pipes designed for use in under-drainage of surface and sub-surface land/ farms. ▪ Canal. Highways. Roads. sports fields, building foundation, construction Sites, Interceptor drainage, In ▪ fields/farms, intercepting canal seepage and under-drainage of hoed canals etc.
	IS 12818: 2010 (Reaffirmed Year: 2021) - Plasticized polyvinyl chloride (PVC - U) screen and casing pipes for bore/tube wells - Specification	▪ covers the requirements of ribbed screen, plain screen and plain casing pipes of nominal diameter 35 mm to 400 mm ▪ produced from un-plasticized polyvinyl chloride for bore/tube-wells for water supply
	IS 14787: 2000(Reaffirmed Year: 2020) - Plasticized PVC pipes (Ducts) and fittings for underground telecommunications cable installation - Specification	▪ Specification and requirement of UPVC pipes used for underground cabling ▪ Covers the requirement of un-plasticized polyvinyl chloride (UPVC) pipes and fitting for use as cable ducts underground telecommunication cable installation

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**ISO 3127: 1994** - Thermoplastics pipes — Determination of resistance to external blows — Round-the-clock method	▪ Specifies a method for the determination of the resistance to external blows of thermoplastics pipes of circular cross-section (round-the-clock method) ▪ This method is applicable to isolated batches of pipe tested at 0 °C.
	ASTM D3835 - Standard test method for determination of properties of polymeric materials by means of a capillary rheometer	▪ Covers measurement of the rheological properties of polymeric materials at various temperatures and shear rates common to processing equipment ▪ measurement of melt viscosity, sensitivity, or stability of melt viscosity with respect to temperature and polymer
	AS/NZS 4765: 2017 - Modified PVC (PVC-M) pipes for pressure applications	▪ This Standard specifies requirements for pipes, integral joints and post-formed bends of PVC-M for the conveyance of water and wastewater under pressure. ▪ The pipes are intended for installation below ground, and above ground where they are not exposed to direct sunlight
	GSO ISO 6259-2: 2015 - Thermoplastics pipes - Determination of tensile properties - Part 2: Pipes made of un-plasticized poly(vinyl chloride) (PVC-U), chlorinated poly (vinyl chloride) (PVC-C) and high-impact poly (vinyl chloride) (PVC-HI)	▪ Specifies a method for determining the tensile properties of pipes ▪ The stress at yield and stress at break ▪ The elongation at break

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 13360: Part 1 : 1992 (Reaffirmed Year: 2018)** - Plastics - Methods of Testing, Part 1: Introduction	▪ Methods for testing. It covers standard atmospheres for conditioning and testing plastics
	IS 4669: 1968 (Reaffirmed Year: 2018): Methods of test for polyvinyl chloride resins	▪ Prescribes methods of test for polyvinyl chloride resins

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**ASTM D2583:** - Standard test method for indentation hardness of rigid plastics by means of a barcol impressor	▪ Method covers the determination of indentation hardness of both reinforced and non-reinforced rigid plastics.
	ASTM F441 - Standard specification for chlorinated polyvinyl chloride (CPVC) plastic pipe	▪ Specification covers chlorinated poly (vinyl chloride) (CPVC) pipe made in Schedule 40 and 80 sizes and pressure-rated for water ▪ Include are criteria for classifying CPVC plastic pipe materials and CPVC plastic pipe

Stage	Code	Scope
	AS/NZS 1462.19-2006 - Methods of test for plastics pipes and fittings, Method 19: C-ring test for fracture toughness of PVC pipes	▪ Specifies a method for establishing the minimum fracture toughness of un-plasticized (vinyl chloride) (PVC-U) pressure pipes
	GSO ISO 4439: 2015 - Plasticized polyvinyl chloride (PVC) pipes and fittings - Determination and specification of density	▪ Specifies a method for the determination of density of pipes and fittings of un-plasticized polyvinyl chloride (PVC) and lays down the permissible limits for this density ▪ The material from which the pipes and fittings are made is essentially polyvinyl chloride to which only the additives required for their manufacture have been added

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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3. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
4. ISO Standards: https://www.iso.org/obp/ui#search
5. EU: https://www.en-standard.eu/
6. USA: https://www.en-standard.eu/astm-standards/, https://www.astm.org/
7. AUS: https://www.standards.org.au/
8. GULF: https://www.gso.org.sa/store/
9. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in
10. Domestic Legislation/Regulation: https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
11. International Legislation/Regulation:
    a. EU: https://ec.europa.eu/growth/sectors/chemicals/reach_en
    b. USA: https://www.cpsc.gov/Regulations-Laws-Standards/Statutes/Summary-List/Consumer-Product-Safety-Improvement-Act-CPSIA, https://www.cpsc.gov/Business-Manufacturing/Business-Education/Business-Guidance/FHSA-Requirements
    c. AUS: https://www.abcb.gov.au/
    d. GULF (GCC): https://www.saso.gov.sa/Documents/SASO-Technical-Regulations-English.pdf

TARPAULINS, KANPUR DEHAT

1. INTRODUCTION

A tarpaulin or tarp is a large sheet of strong, flexible, water-resistant or waterproof material such as canvas or polyester coated with polyurethane, or made of plastics such as polyethylene.

Tarpaulins often have reinforced grommets at the corners and along the sides to form attachment points for rope, allowing them to be tied down or suspended. Inexpensive modern tarpaulins are made from woven polyethylene; this material is so associated with tarpaulins that it has become colloquially known in some quarters as polytarp. Plastic goods industry is rapidly growing in Kanpur Dehat which is a major hub for the production of tarpaulins.

The word tarpaulin originated as a compound of the words tar and palling, referring to a tarred canvas pall used to cover objects on ships. Sailors often tarred their own jackets worn over their garments in the same manner as the sheets or palls. Tarpaulins are large water-resistant sheets that provide temporary waterproofing. Reinforced metal eyelets along the seams are used with tarpaulin ties, ropes or bungees to secure the tarp in place. Tarpaulins are used in many ways to protect persons and things from wind, rain, and sunlight. They are used during construction or after disasters to protect partially built or damaged structures, to prevent mess during painting or similar activities and to contain and collect debris. High-density polyethylene (HDPE) tarpaulin has a high demand in India for domestic, commercial as well as industrial use. HDPE tarpaulin is a polymer having multiple end uses and is considered a boon for the Indian industry. It's excellent for covering purposes, helping protect products from dust and moisture. 425 GSM tarpaulins are very strong & durable to protect material from heavy rain & wind.

Tarpaulins can be classified based on a diversity of factors, such as material type (polyethylene, canvas, vinyl, etc.), thickness, which is generally measured in mils or generalized into categories (such as "regular duty", "heavy duty", "Super heavy duty", etc.), and grommet strength (simple vs. reinforced), among others. A polyethylene tarpaulin ("*Polytarp*") is not a traditional fabric, but rather, a laminate of woven and sheet material. Sheets can be either of low density polyethylene (LDPE) or high density polyethylene (HDPE). When treated against ultraviolet light, these tarpaulins can last for years exposed to the elements, but non-UV treated material will quickly become brittle and lose strength and water resistance if exposed to sunlight over prolonged periods. HDPE Tarpaulins are light-weight, easy to handle, water proof and can be manufactured in any desired colours.

2. RAW MATERIALS

Tarpaulins can be manufactured out of a wide variety of materials such as canvas, vinyl (PVC), polyethylene and HDPE (high density poly-ethylene). However, HDPE is largely used.

3. MANUFACTURING PROCESS FOR POLYETHYLENE TARPAULINS

The manufacturing process of plastic tarpaulins can be described as follows:

3.1. Core

Most tarpaulins start off as polyethylene in pellet form. These HDPE pellets are fed into an extruding line which first melts the pellets and produces polyethylene film, and then cuts and stretches the film into polyethylene yarn. A weaving loom then cross-weaves the yarn into a fabric that is tear-resistant and stretch-resistant.

3.2. Lamination

The HDPE fabric is laminated with LDPE (Low Density Polyethylene)/ LLDPE (Linear Low Density Polyethylene) or a blend of the two. A tarpaulin could also be a three-layer tarpaulin having one layer of woven fabric and two layers of LDPE/LLDPE coating, one each on either side. Five layer tarpaulins consisting of two layers of woven fabric sandwiched between layers of LDPE/LLDPE coating are also made for special uses such as camping etc.

3.3. Border Making

A border is made and a rope is provided along the border to provide strength. Metallic loops are used to make eyelets along the border, through which the fastening ropes are passed.

4. GENERAL QUALITY CHECKS

Following are a few quick visual quality checks:

- Check for production defects, such as bumps, depressions, chipping and tear
- Check the weight of the assembled product and GSM (Gram/square meter)
- The colour of the material should be uniform
- Verify the measurements of an assembled product follow contract specifications.

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European

Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	NA	NA	NA

5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	European Union	a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations (EC) 1907/2006	a. European Commission b. European Chemicals Agency (ECHA)	a. Hazardous chemicals b. Persistent organic pollutants c. General product safety directive
2	United States of America	a. Consumer Product Safety Improvement Act (CPSIA), Act of 2008 b. Consumer Product Safety Act (CPSA) Act of 1972 c. Federal Hazardous Substances Act (FHSA), Act of 1960 d. California Proposition 65, Act of 1986	a. Consumer Product Safety Commission (CPSC) b. California State	a. Regulations on chemicals b. Quantity of regulated material c. Product labelling d. Surface flammability e. Hazardous chemicals
3	Middle East	a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations b. SASO Certificate of Conformity	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. General product safety regulation b. Labelling Requirement

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 2828: 2019**	Plastics — Vocabulary (Second Revision)	Middle East	**GSO ISO 21067: 2010**	Vocabulary of packaging
	IS 2244: 2017 (Reaffirmed Year: 2017)	Glossary of terms relating to treated fabrics	-	-	-
Raw Materials (Refer Table	**IS 2530: 1963 (Reaffirmed Year: 2018)**	Methods of test of materials	-	-	-
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 13360: Part 1: 1992: (Reaffirmed Year: 2018)**	Method of testing for plastic	European Union	**ISO 293: 2004**	Compression moulded test specimens of thermoplastics materials
	IS 7903: 2017	Textiles - Tarpaulins made from high density polyethylene (HDPE) woven fabrics - specification (fifth revision)	European Union	**ISO 178: 1975**	Flexural properties of rigid plastics
	IS 6899: 1997 (Reaffirmed Year: 2017)	Requirement of fabric for manufacturing tarpaulins	Australia	**AS 3520**	Specifications and details of tarpaulins
	IS 7941: 1976 (Reaffirmed Year: 2020)	Method for determining water repellency	Middle East	**GSO 1951: 2009**	Details of fabrics used to manufacture tarpaulins
	IS 2089: 1977 (Reaffirmed Year: 2017)	Specification for common proofed canvas/duck and paulins (tarpaulins)	Australia	**AS 2930**	Coated fabric for tarpaulins
	-	-	European Union	**ISO 1183 - 3: 1999**	Density of solid plastics
Finishing (Refer Table 4.a & 4.b)	**IS 2789: 1972 (Reaffirmed Year: 2017)**	Specification for special proofed paulins (tarpaulins)	Middle East	**GSO ISO 811: 2021**	Determination of resistance to water penetration
	IS 14611: 2016	Multi-layered cross laminated sheets/ tarpaulins/ covers/ agricultural films specification	USA	**ASTM D3835**	Rheological properties of thermoplastics
	-	-	USA	**ASTM D2583**	Hardness of plastics
	-	-	USA	**ASTM D751**	Methods of test for coated fabrics

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 2828: 2019** - Plastics – Vocabulary	▪ Define the terms used in plastic industry, English and French ▪ Synonymous terms follow the preferred term
	IS 2244: 2017 (Reaffirmed Year: 2017) - Glossary of terms relating to treated fabrics	▪ Glossary of terms
	GSO ISO 21067 - 2010 - Packaging – Vocabulary	▪ Specifies preferred terms and definitions related to packaging and materials handling, for use in international commerce

6.2. Table 2: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 2530: 1963 (Reaffirmed Year: 2018)** - Methods of test for polyethylene moulding materials and polyethylene compounds	▪ Prescribes the methods of test for polyethylene moulding materials and polyethylene compounds ▪ Some are for Apparent Density, Bulk Factor, Elongation at Break and Melt Flow

6.3. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 13360: Part 1 : 1992 (Reaffirmed Year: 2018)** - Plastics - Methods of testing, Part 1: Introduction	▪ Methods for testing. It covers standard atmospheres for conditioning and testing plastics
	IS 7903: 2017 - Textiles - Tarpaulins made from high density polyethylene woven fabric	▪ Prescribes constructional and other requirements for tarpaulins made from high density polyethylene woven fabric having minimum mass of 200 g/m2
	IS 6899: 1997 (Reaffirmed Year: 2017) - Textiles - High density polyethylene (HDPE) woven fabrics	▪ This standard specifies requirements of five varieties of high density polyethylene (HDPE) ▪ Woven fabrics generally used for wrapping, bale covering and similar applications
	IS 7941: 1976 (Reaffirmed Year: 2020) - Method for determining the water repellency of fabrics by cone test	▪ Prescribes a method for determining the water repellency of fabrics ▪ Generally applicable for testing heavier types of proofed fabrics used as covers for wagons, shelters, etc.
	IS 2089: 1977 (Reaffirmed Year: 2017) - Specification for common proofed canvas/duck and paulins (tarpaulins)	▪ Specifications for use of copper or zinc napthenate as an anti-rot proofing agent for tarpaulins ▪ Declarations to be made ▪ Adjunct standards

6.4. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**ISO 293: 2004** - Plastics — Compression moulding of test specimens of thermoplastic materials	▪ Specifies the general principles and the procedures to be followed with thermoplastics in the preparation of compression-moulded test specimens ▪ The main steps of the procedure, including four different cooling methods, are standardized ▪ The required moulding temperature and cooling methods are as specified in the appropriate International Standard for the material or as agreed between the interested parties

Stage	Code	Scope
	ISO 178: 1975 - Plastics — Determination of flexural properties	▪ A method for determining the flexural properties of rigid and semi-rigid plastics under defined conditions ▪ Thermoplastic moulding, extrusion and casting materials, including filled and reinforced compounds in addition to unfilled types
	ISO 527-2: 2012 - Plastics — Determination of tensile properties	▪ Specifies the conditions for determining the tensile properties of plastics films or sheets less than 1 mm thick, based upon the general principles
	AS 3520 - Textiles — Tarpaulins	▪ Specifies requirements for tarpaulins and shaped covers including canopies manufactured from: ▪ Classifies tarpaulins and shapes covers into two types depending on the material they are made from, and into four duties ▪ Covers seams, stitching, and appurtenances such as shortening lugs, eyelets and ropes
	GSO 1951: 2009 - PVC-coated fabrics for tarpaulins	▪ Concerned with fabric coated on one or both sides with a plasticized coating, pigmented or otherwise, of polyvinyl chloride (PVC) or copolymer ▪ The major constituent of which is Vinyl chloride and which is suitable for use in the making-up of tarpaulins
	AS 2930 -Textiles — Coated fabrics for tarpaulins	▪ Specifies the requirements for coated fabrics using natural or synthetic fibres or their blends ▪ Either in woven or knitted construction, suitable for manufacture of tarpaulins

6.5. Table 4.a: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 2789: 1972 (Reaffirmed Year: 2017)** - Specification for special proofed paulins (tarpaulins)	▪ Requirements and methods of sampling ▪ Tests for specially proofed tarpaulins ▪ Referred documents and adjunct standards
	IS 14611: 2016 - Multi-layered cross laminated sheets/ tarpaulins/ covers/ agricultural films specification	▪ Construction details ▪ Methods for sampling and tests ▪ Referenced documents

6.6. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**ISO 1183 - 3: 1999** - Plastics — Methods for determining the density and relative density of non-cellular plastics	▪ Specification of four methods for the determination of the density of plastics except cellular plastics ▪ Immersion method for plastics in a finished condition ▪ Pyknometer method for plastics in the form of small particles such as powders, granules pellets or flakes
	GSO ISO 811: 2021 - Textile fabrics - Determination of resistance to water penetration - Hydrostatic pressure test	▪ Specifies a hydrostatic pressure method for determining the resistance of fabrics to penetration by water ▪ The method is primarily intended for dense fabrics, e.g. ducks, tarpaulins and tenting's
	ASTM D3835 - Standard Test Method for Determination of Properties of Polymeric Materials by Means of a Capillary Rheometer	▪ Covers measurement of the rheological properties of polymeric materials at various temperatures and shear rates common to processing equipment ▪ measurement of melt viscosity, sensitivity, or stability of melt viscosity with respect to temperature and polymer

Stage	Code	Scope
	ASTM D2583 - 13a - Standard Test Method for Indentation Hardness of Rigid Plastics by Means of a Barcol Impressor	▪ Method covers the determination of indentation hardness of both reinforced and non-reinforced rigid plastics
	ASTM D751 - Standard Test Methods for Coated Fabrics	▪ Test methods cover, but are not limited to, rubber-coated fabrics, that is, tarpaulins, rainwear, and similar products

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

1. Product information- http://odopup.in/en/article/kanpur-dehat
2. Domestic Standards and raw materials: https://standardsbis.bsbedge.com/
3. ISO Standards: https://www.iso.org/obp/ui#search
4. EU: https://www.en-standard.eu/
5. USA: https://www.en-standard.eu/astm-standards/, https://www.astm.org/
6. AUS: https://www.standards.org.au/
7. GULF: https://www.gso.org.sa/store/
8. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in
9. Domestic Legislation/Regulation:: https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
10. International Legislation/Regulation:
    a. EU: https://ec.europa.eu/growth/sectors/chemicals/reach_en
    b. USA: https://www.cpsc.gov/Regulations-Laws-Standards/Statutes/Summary-List/Consumer-Product-Safety-Improvement-Act-CPSIA; https://www.cpsc.gov/Business-Manufacturing/Business-Education/Business-Guidance/FHSA-Requirements
    c. AUS: https://www.abcb.gov.au/
    d. GULF (GCC): https://www.saso.gov.sa/Documents/SASO-Technical-Regulations-English.pdf

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials, moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/ calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL)**: https://nabl-Indiaorg/
- **Bureau of Indian Standards (BIS)**: https://bis.gov.in/

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB): http://nabcb.qci.org.in/

APPENDIX II: IMPORTANT INSTITUTIONS

PLASTICS				
S. No.	**Institute**	**Areas of Expertise**	**Postal Address**	**Contact**
1	CSIR-Indian Institute of Toxicology Research	▪ Consultancy ▪ Transfer of Technology ▪ Development of Technology	CSIR-Indian Institute of Toxicology Research, Vishvigyan Bhawan, 31, Mahatma Gandhi Marg, Lucknow - 226 001. Uttar Pradesh, India	Email: director@iitrindia.org Phone: +91-522-2217497 Website: http://iitrindia.org/
2	CIPET-Institute of Petrochemicals Technology	▪ Skill Training ▪ Technology Support	CIPET: Institute of Petrochemicals Technology (IPT), B-27, Amausi Industrial Area, Lucknow - 226 008	Email: lucknow@cipet.gov.in Phone: +91-522-2436227 Website: https://www.cipet.gov.in/

ONE DISTRICT ONE PRODUCT

UTTAR PRADESH